RHEOLOGICAL BIOMARKERS FROM BROADBAND TORSIONAL WAVE ELASTOGRAPHY FOR TUMOUR DIAGNOSIS

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Introduction

Clinical elastography still reports tumours through a single stiffness value, yet malignant stromal remodelling also alters viscosity, relaxation behaviour and mechanical anisotropy. Benign and malignant lesions can share a similar shear modulus while differing markedly in frequency-dependent dissipation, so stiffness alone may under-diagnose ambiguous lesions. Torsional wave elastography (TWE) generates and detects shear waves over a broader frequency range than commercial systems with the spatial resolution required for superficial tumours, enabling reconstruction of the relaxation spectrum linked to collagen remodelling, cellular density changes and interstitial fluid redistribution around a tumour. This work investigates whether broadband, model-free rheological spectra discriminate malignant from benign tissue beyond stiffness.

Methods

A TWE platform excites torsional shear waves across a broadband spectrum and measures the local shear wave speed $c_s(\omega)$ at each frequency. A model-free relaxation spectrum $H(\tau)$ is reconstructed directly from the measured dispersion curve, without assuming a Kelvin-Voigt, Maxwell or fractional law beforehand:

$$H(\tau) \approx (\pi/2)\cdot\rho\cdot c_s^2(1/\tau)\cdot(d \ln c_s(\omega) / d \ln \omega)\,|_{\omega=1/\tau} \quad (1)$$

relaxation times τ_j and weights $H(\tau_j)$ are the extrema of this spectrum. The probe was benchmarked against two reference platforms, SWE and TW-UI, on tissue-mimicking phantoms (patent process IPR-1092), then applied ex vivo in animal tissue (skin, liver, fat, muscle) and in a clinical pilot on patients with cutaneous lesions scheduled for excision, scanning lesion and control skin with TWE before surgery and grading specimens histopathologically, from naevus to carcinoma and melanoma.

Results

In animal tissue, broadband TWE consistently resolved two to three well-separated relaxation peaks per sample across skin, liver, fat and muscle, spanning sub-millisecond to several-millisecond time scales that a single average shear velocity cannot resolve. In the skin-cancer pilot, dispersion curves and reconstructed spectra differed between excised lesions and adjacent healthy skin, and the resulting mechanical biomarker tracked the histopathological malignancy score across patients, from naevus to melanoma.

Conclusions

Reformulating elastography around a relaxation spectrum, rather than a single stiffness value, adds a rheological dimension sensitive to tumour-induced histological remodelling. Broadband torsional wave elastography with model-free spectral reconstruction yields mechanical biomarkers with promising discriminative value for skin tumour diagnosis. Larger cohorts and quantitative histopathological correlation are still required, but these results position tissue rheology as a candidate tumour biomarker complementing stiffness-based elastography.

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References

Rus G, Faris IH, Torres J, Callejas A, Melchor J. Sensors, 20:2379, 2020.
Almashakbeh Y, Shamimi H, Callejas A, Rus G. Scientific Reports, 14:16058, 2024.